Anirban Nath

+1-332-251-8609 | an3145@columbia.edu | linkedin.com/in/anirban7733

Education

Columbia University, New York, NY 2022 – Present
Ph.D. in Statistics, (GPA: 3.9/4.0)

Indian Statistical Institute, Kolkata, India
Master of Statistics, First Class with Distinction 2020 – 2022
Bachelor of Statistics, First Class with Distinction 2017 – 2020

Experience

Data Science Intern, Capital One Summer 2026

- Developed a causal inference framework to predict heterogeneous treatment effects, optimizing for incremental lead generation per dollar of subscription spend.
- Built a Causal-ML targeting pipeline to rank ∼10K dealer prospects, capturing 60% of estimated incremental lead opportunity and achieving 70% similarity to the benchmark ranking.

Research

- **Weighted Conformal Clustering.** With Y Hur and G Allen. arXiv:2606.00436 NeurIPS, 2026.
- **Inference for Clustering: Conformal Sets for Cluster Labels.** With Y Hur and G Allen. arXiv:2604.03488 Submitted to the Journal of the American Statistical Association.
- **Linear Regression Beyond Covariance: Latent Slope Spectra from Higher-Order Cumulants** With M Yuan. Submitted to Biometrika.
- **Inference for Portfolio Optimization in High Dimensions.** With M Yuan. (In preparation)
- **Concentration of Adjacency and Laplacian Matrices for Lazy Network-Valued Stochastic Processes.** With S Chatterjee et al. arXiv:2208.01365.
- **A Bayesian Predictive Modeling Framework for Bivariate Count Data and Application in Forecasting Soccer Matches.** With S Deb. (In preparation)
- **Differential Fault Analysis of NORX Using Variants of the Coupon Collector Problem.** With A Jana et al. (**Published** in Journal of Cryptographic Engineering, Springer)
- **Modeling Tips of Yellow Cab Drivers in New York City.** With S Laha and S Ghosh.

Technical Skills

- **Programming:** Python, R, SQL, C, LaTeX.
- **Libraries:** pandas, scikit-learn, statsmodels, Keras, TensorFlow, XGBoost, LightGBM, CausalML.
- **Data and infrastructure:** Snowflake, AWS, Git, Linux/bash.
- **Spoken languages:** English, Bengali, Hindi.

Awards and Honors

- Stonehage-Fleming Prize for the **3rd highest score in India** in the Simon Marais Mathematics Competition.
- Merit scholarships for academic excellence in **four consecutive years** at Indian Statistical Institute, Kolkata.
- Selected among the **top 25 students from India** in the Madhava Mathematics Competition.
- Ranked **8** out of **20K** in ISI Entrance Examination all over India.

Leadership and Service

- Student representative, Department of Statistics, Columbia University, 2025.
- Mentor for the Inference qualifying examination, 2024 and 2025.
- Mentor for the Core Competency examination, 2024.
- Student seminar organizer, 2023–2024.

Other Interests

Contract Bridge (USBF Junior), Playing card collection, Card magic, and Table tennis.